

Supplementary Material

Radiomics-Based Machine Learning and Deep Learning for Clinically Significant Prostate Cancer Detection on Biparametric MRI

S1 Sensitivity analysis: optimistic upper bound for the deep models under test-fold epoch selection

A natural concern when classical and deep models are compared is whether the deep architectures were disadvantaged by their training protocol. In the main analysis, the number of training epochs for each deep model was selected on an internal 80:20 split of the outer-training partition, after which the model was refit on the complete outer-training partition; the held-out outer fold was never used for any fitting or model-selection decision, exactly mirroring the leakage-safe treatment of the classical ensembles. To bound how much the deep models could gain from a more permissive—but optimistically biased—protocol, we repeated the deep-model experiments while selecting the training epoch by monitoring the held-out outer fold itself (up to 300 epochs). This procedure deliberately leaks information from the evaluation fold into model selection and therefore yields an *upper bound* on deep-model performance rather than a valid estimate of generalization; the classical models were left unchanged and serve as the leakage-safe reference. All metrics are pooled out-of-fold values with patient-level cluster-bootstrap 95% confidence intervals, reported at the fixed 0.5 probability threshold.

Even under this favorable protocol, the deep models did not overturn the main conclusions (Table S1). In the radiomics-only setting—the primary comparison—the leakage-safe Random Forest (AUROC 0.814) remained at least as discriminative as every test-fold-selected deep model (best deep AUROC 0.794), so the classical ensemble retained its advantage despite the deep models being granted a peek at the evaluation fold. Only in the combined conditions did the optimistically tuned deep models edge ahead of the Random Forest, and then only marginally and with broadly overlapping confidence intervals: by +0.009 AUROC for the concatenation Transformer (0.837 vs. 0.828) and by +0.015 AUROC for the dual-branch Transformer (0.843 vs. 0.828). These small, leakage-inflated margins are consistent with the main finding that, on this engineered radiomic feature space, deep tabular models do not confer a robust discrimination advantage over a calibrated tree ensemble, and they reinforce rather than weaken that conclusion: the deep models fail

to clearly surpass the classical baseline even when the evaluation protocol is tilted in their favor.

Table S1: Optimistic upper bound for the deep models when the training epoch is selected on the held-out outer fold (a deliberately leakage-prone protocol), compared with the leakage-safe Random Forest reference carried over unchanged from the main analysis. Threshold-free metrics on pooled out-of-fold predictions with patient-level cluster-bootstrap 95% confidence intervals; lower Brier is better. Deep-model rows are optimistic upper bounds and are *not* valid estimates of generalization.

Condition	Model (protocol)	AUROC	AUPRC	Brier
Radiomics-only	Random Forest (leakage-safe, reference)	0.814 (0.790–0.837)	0.605 (0.561–0.655)	0.152
	Transformer (test-fold epoch)	0.794 (0.766–0.819)	0.585 (0.544–0.636)	0.161
	CapsNet (test-fold epoch)	0.761 (0.735–0.788)	0.559 (0.519–0.608)	0.202
	Transformer–CapsNet (test-fold epoch)	0.793 (0.767–0.818)	0.588 (0.545–0.634)	0.177
Radiomics+clinical (concat)	Random Forest (leakage-safe, reference)	0.828 (0.805–0.851)	0.634 (0.591–0.683)	0.147
	Transformer (test-fold epoch)	0.837 (0.815–0.860)	0.649 (0.606–0.697)	0.154
	CapsNet (test-fold epoch)	0.831 (0.809–0.853)	0.646 (0.605–0.691)	0.170
	Transformer–CapsNet (test-fold epoch)	0.828 (0.806–0.852)	0.652 (0.612–0.696)	0.166
Radiomics+clinical (dual)	Random Forest (leakage-safe, reference)	0.828 (0.805–0.851)	0.634 (0.591–0.683)	0.147
	Dual Transformer (test-fold epoch)	0.843 (0.821–0.864)	0.671 (0.628–0.716)	0.154
	Dual CapsNet (test-fold epoch)	0.840 (0.817–0.861)	0.661 (0.620–0.706)	0.151
	Dual Transformer–CapsNet (test-fold epoch)	0.829 (0.806–0.851)	0.621 (0.581–0.673)	0.171

S2 Radiomics-only per-model operating-point performance

Table S2 reports the full set of discrimination, calibration, and operating-point metrics for all seven radiomics-only models, complementing the forest plot of AUROC and AUPRC in the main text. Threshold-free metrics (AUROC, AUPRC, Brier) use the calibrated probabilities; sensitivity, specificity, balanced accuracy, F1, PPV, and NPV are reported at the per-fold outer-training-derived Youden operating point. Values are point estimates with patient-level cluster-bootstrap 95% confidence intervals for AUROC, AUPRC, sensitivity, and specificity (lower Brier is better).

Table S2: Radiomics-only model performance on pooled out-of-fold predictions.

Model	AUROC	AUPRC	Brier	Sensitivity	Specificity	Bal. acc.	F1	PPV	NPV
TabFM (foundation)	0.828 (0.805–0.850)	0.650 (0.600–0.699)	0.145	0.812 (0.774–0.848)	0.691 (0.664–0.718)	0.751	0.626	0.510	0.903
Random Forest (ML)	0.814 (0.790–0.837)	0.605 (0.556–0.658)	0.152	0.767 (0.724–0.806)	0.736 (0.710–0.761)	0.751	0.630	0.534	0.889
Gradient Boosting (ML)	0.802 (0.777–0.825)	0.592 (0.542–0.644)	0.156	0.774 (0.733–0.813)	0.675 (0.647–0.703)	0.725	0.597	0.485	0.883
Transformer (DL)	0.801 (0.775–0.824)	0.609 (0.559–0.662)	0.155	0.755 (0.714–0.795)	0.684 (0.657–0.711)	0.720	0.591	0.486	0.876
Transformer–CapsNet (DL)	0.803 (0.777–0.826)	0.600 (0.550–0.651)	0.154	0.807 (0.768–0.843)	0.672 (0.644–0.699)	0.739	0.612	0.493	0.898
LightGBM (ML)	0.792 (0.767–0.817)	0.598 (0.548–0.650)	0.159	0.791 (0.751–0.830)	0.672 (0.644–0.700)	0.731	0.603	0.488	0.890
CapsNet (DL)	0.794 (0.768–0.820)	0.589 (0.539–0.643)	0.155	0.753 (0.710–0.793)	0.685 (0.658–0.712)	0.719	0.590	0.486	0.875

S3 Model-agnostic SHAP attributions for the pretrained TabFM model

Because TabFM performs inference by in-context learning rather than through a differentiable forward pass over fixed weights, gradient-based attribution methods such as integrated gradients

are not applicable. To characterize which features drove its predictions, model-agnostic permutation SHAP was instead computed for the pretrained TabFM model in the radiomics-only, clinical-only, and concatenation conditions. For each of the five outer folds, SHAP values were estimated on a held-out subsample of 25 validation studies against a 50-study background set, and the resulting per-study signed attributions were pooled across folds (Figure S1). The subsample size makes these attributions qualitative rather than a basis for fine-grained ranking, but they were computed with an attribution method independent of those used for the classical (SHAP) and deep (integrated gradient) models and therefore provide a cross-check on the shared feature signature. TabFM’s leading features closely matched those of the other model families: diffusion-weighted texture and ADC first-order total energy dominated the radiomics-only condition, PSA density dominated the clinical-only condition, and in the concatenation condition PSA density and age rose above every radiomic descriptor—reproducing the re-ranking seen for the classical and deep combined models and confirming that the clinical variables supply largely independent risk information.

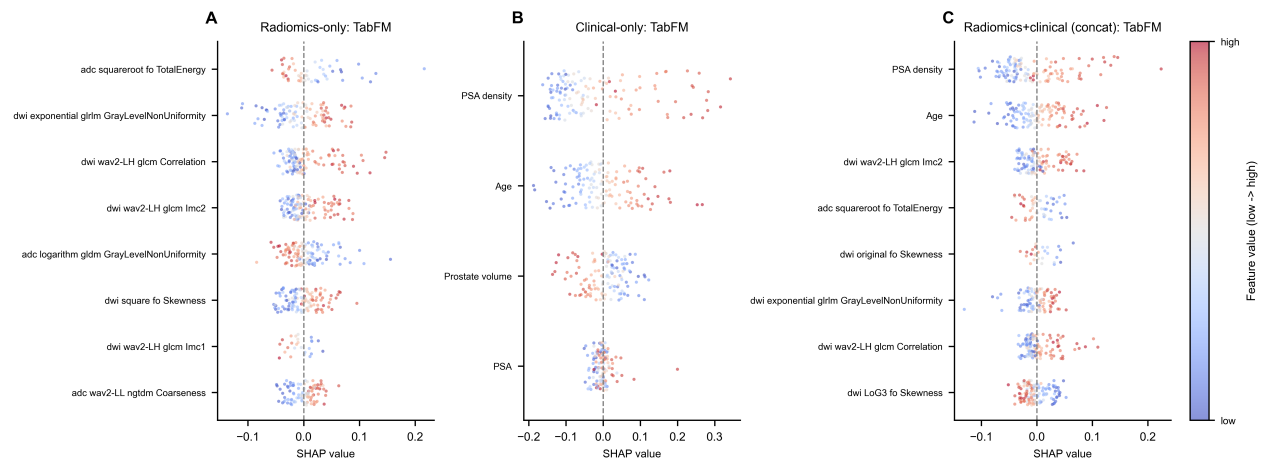


Figure S1: Model-agnostic SHAP attributions for the pretrained TabFM model. Beeswarm summaries of per-study signed permutation-SHAP values, pooled across the five outer folds’ held-out subsamples, for the eight most important features in the (A) radiomics-only, (B) clinical-only, and (C) concatenation conditions. Each point is a study, positioned by its signed SHAP value and colored by the relative feature value (low to high); values to the right of zero increase the predicted csPCa probability. PSA density is the leading feature in the clinical-only and concatenation conditions, and diffusion-weighted texture with ADC first-order total energy leads the radiomics-only condition, matching the signature recovered by the classical and deep models with independent attribution methods.

S4 Voxel-based radiomic feature maps

To provide case-level spatial explanations directly on the anatomy, a selected set of influential features was back-projected to the image as voxel-wise radiomic maps in csPCa-positive cases with human-expert lesion delineations (Figure S2). Their salient regions co-localized with the delineated csPCa lesion rather than being distributed uniformly through the gland, complementing the global importance summaries in the main text.

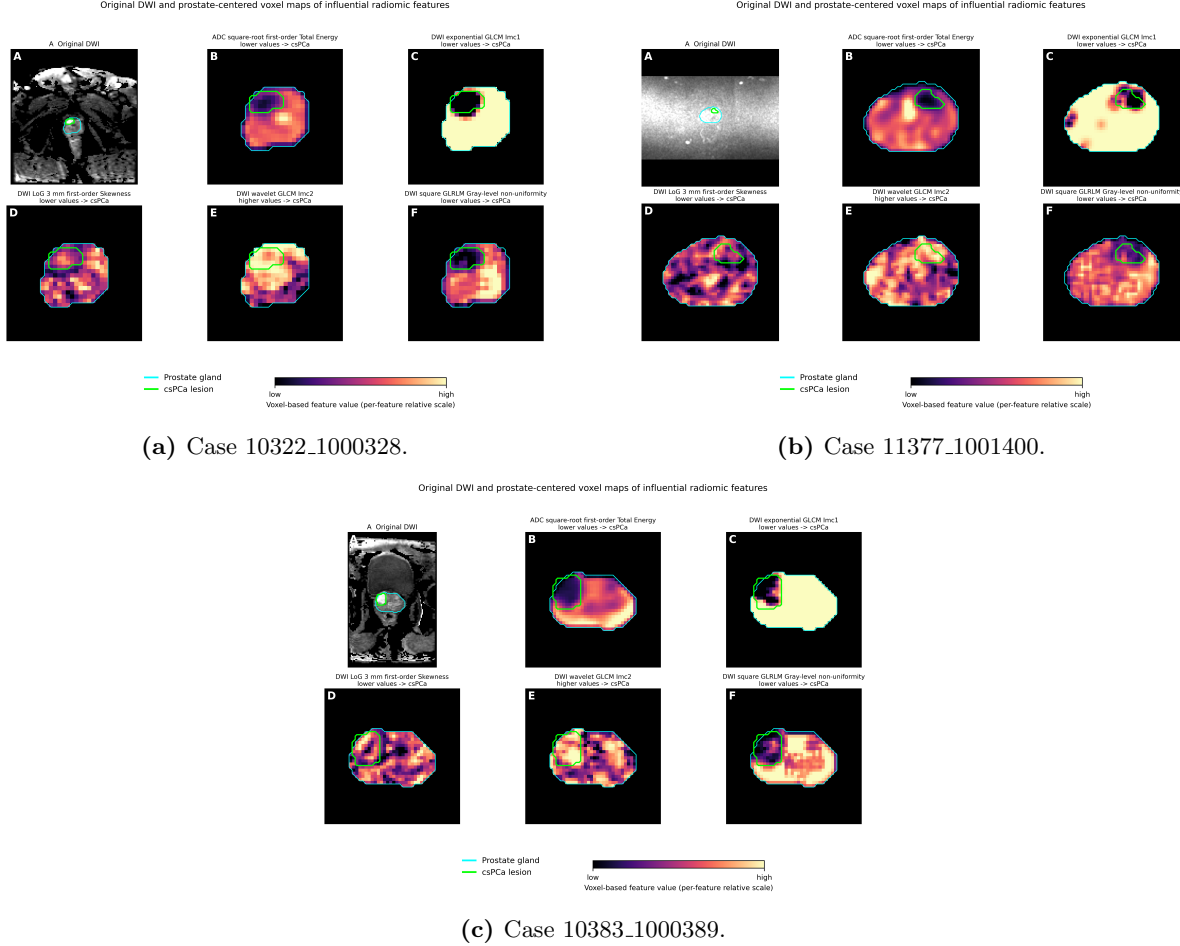


Figure S2: Original DWI and prostate-centered voxel-based radiomic maps in csPCa-positive cases with human-expert lesion delineations. In each subfigure, (A) original DWI slice with prostate gland (cyan) and csPCa lesion (green) delineations only; (B) ADC square-root first-order total energy, (C) DWI exponential GLCM Imc1, (D) DWI LoG 3 mm first-order skewness, (E) DWI wavelet GLCM Imc2, and (F) DWI square GLRLM gray-level non-uniformity. Feature panels show only the voxel map within a prostate-centered bounding box, with the gland and lesion contours retained. Colour encodes voxel-based feature value on a per-feature relative scale (low to high). Panel titles indicate the feature direction from the signed SHAP/integrated-gradient beeswarm analyses: lower values support the positive csPCa class for B–D and F, whereas higher values support the positive csPCa class for E.

S5 Cross-center transportability (leave-one-center-out)

Table S3 reports the full leave-one-center-out (LOCO) results for every classical and TabFM model in each condition (the deep models were not re-run under LOCO). In each of the three LOCO folds the model was trained on two centers and tested on the entirely held-out third center, and predictions were pooled over the three held-out centers; all feature selection, calibration, and (for TabFM) in-context background rows were confined to the training centers. Discrimination fell by roughly 0.10–0.14 AUROC relative to the center-mixed five-fold estimates for every model family, with TabFM retaining a small numerical lead in every condition and its relative advantage largest in the combined condition. Pooled LOCO AUPRC for TabFM was 0.461 (radiomics-only), 0.480 (clinical-only), and 0.536 (concatenation).

Table S3: Leave-one-center-out transportability: pooled out-of-fold AUROC over the three held-out centers with patient-level cluster-bootstrap 95% confidence intervals. “Five-fold” repeats the center-mixed primary estimate for reference.

Condition	Model	LOCO AUROC (95% CI)	Five-fold AUROC
Radiomics-only	Random Forest	0.676 (0.647–0.706)	0.814
	Gradient Boosting	0.694 (0.663–0.723)	0.802
	LightGBM	0.695 (0.663–0.725)	0.792
	TabFM	0.702 (0.674–0.729)	0.828
Clinical-only	Random Forest	0.697 (0.666–0.729)	—
	Gradient Boosting	0.700 (0.668–0.731)	—
	LightGBM	0.676 (0.644–0.709)	—
	TabFM	0.718 (0.690–0.746)	0.757
Radiomics+clinical (concat)	Random Forest	0.708 (0.678–0.738)	0.828
	Gradient Boosting	0.728 (0.699–0.756)	—
	LightGBM	0.708 (0.678–0.739)	—
	TabFM	0.753 (0.726–0.779)	0.844

S6 PSA-density source-complete sensitivity

To confirm that PSA-density handling did not drive the results, all conditions were re-run on the source-complete cohort of 1,049 studies with an originally recorded (non-derived, non-imputed) PSA density (Table S4). Discrimination was preserved and, if anything, slightly higher than in the full cohort with deterministic derivation and residual median imputation, across all model families. The full-cohort derived-plus-median-imputed analysis ($n = 1500$) likewise matched the primary results (clinical-only Gradient Boosting AUROC 0.748; concatenation Random Forest 0.826).

Table S4: Source-complete PSA-density sensitivity ($n = 1049$): pooled out-of-fold AUROC with patient-level cluster-bootstrap 95% confidence intervals for the classical models; TabFM values are pooled out-of-fold point estimates.

Condition	Model	AUROC (95% CI)
Radiomics-only	Random Forest	0.832 (0.806–0.858)
	Gradient Boosting	0.831 (0.804–0.858)
	LightGBM	0.803 (0.774–0.834)
	TabFM	0.851
Clinical-only	Gradient Boosting	0.775 (0.743–0.806)
	Random Forest	0.771 (0.737–0.804)
	LightGBM	0.749 (0.714–0.783)
	TabFM	0.796
Radiomics+clinical (concat)	Random Forest	0.857 (0.834–0.881)
	Gradient Boosting	0.856 (0.831–0.881)
	LightGBM	0.856 (0.832–0.880)
	TabFM	0.873

S7 Calibration-method sensitivity (isotonic vs Platt)

Replacing Platt scaling with isotonic regression left discrimination and the ranking of the classical models essentially unchanged, as expected for a monotonic recalibration (Table S5); the largest change in AUROC relative to the Platt-calibrated primary analysis was 0.004.

Table S5: Isotonic-calibration sensitivity: pooled out-of-fold AUROC with patient-level cluster-bootstrap 95% confidence intervals (classical models, full cohort $n = 1500$). Platt-calibrated primary values are shown for reference.

Condition	Model	Isotonic AUROC (95% CI)	Platt AUROC
Radiomics-only	Random Forest	0.813 (0.789–0.837)	0.814
	LightGBM	0.800 (0.775–0.825)	0.792
	Gradient Boosting	0.796 (0.770–0.819)	0.802
Radiomics+clinical (concat)	Random Forest	0.824 (0.800–0.846)	0.828
	Gradient Boosting	0.820 (0.796–0.842)	—
	LightGBM	0.816 (0.792–0.841)	—

S8 Feature-selection stability

Fold-specific feature selection was highly reproducible. For the radiomics-only plan a mean of 67.6 features was retained per fold (range 67–69) from 99 distinct features across folds, 41 of which were selected in all five folds; the pairwise Jaccard overlap of the five fold-specific feature sets is shown in Table S6 (mean 0.68). The radiomics+clinical plan was comparably stable (mean pairwise Jaccard 0.69, range 0.61–0.78).

Table S6: Pairwise Jaccard overlap of the five fold-specific selected-feature sets for the radiomics-only plan.

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Fold 1	1.00	0.77	0.63	0.63	0.68
Fold 2	0.77	1.00	0.66	0.67	0.77
Fold 3	0.63	0.66	1.00	0.59	0.65
Fold 4	0.63	0.67	0.59	1.00	0.75
Fold 5	0.68	0.77	0.65	0.75	1.00

S9 Decision-curve biopsy avoidance

Table S7 translates the decision-curve analysis into biopsy-avoidance terms for the best model in each condition, evaluated at its outer-training-derived Youden operating point. Relative to a biopsy-all strategy, the combined models avoided the most biopsies while maintaining a high negative predictive value. The dual-stream late-fusion TabFM operated at a substantially higher probability threshold (0.53), so its biopsy-avoidance count is not directly comparable to the other operating points.

Table S7: Biopsy-avoidance summary at the selected Youden operating point for the best model in each condition. Net benefit and biopsies avoided per 100 patients are relative to a biopsy-all strategy at the same threshold.

Condition	Model	Threshold	Sens.	Spec.	PPV	NPV	Net benefit	Biopsies avoided/100
Clinical-only	TabFM	0.30	0.628	0.749	0.497	0.836	0.100	29.4
Radiomics-only	TabFM	0.22	0.819	0.696	0.516	0.907	0.171	31.5
Radiomics+clinical (concat)	TabFM	0.27	0.793	0.745	0.552	0.901	0.158	37.2
Radiomics+clinical (late fusion)	TabFM	0.53	0.762	0.763	0.560	0.890	0.024	48.7

S10 Per-center performance for all models

Table S8 extends the best-model per-center summary of the main text to every model in every condition, on the primary pooled out-of-fold predictions. Center sizes and prevalence were: RUMC $n = 800$ (29.5% csPCa), ZGT $n = 350$ (22.9%), and PCNN $n = 350$ (31.1%). PCNN was the lowest-performing center for every model and condition.

Table S8: Per-center AUROC on the primary pooled out-of-fold predictions for all models. RUMC, Radboud University Medical Center; ZGT, Ziekenhuisgroep Twente; PCNN, Prostaat Centrum Noord-Nederland.

Condition	Model	RUMC	ZGT	PCNN
Radiomics-only	Gradient Boosting	0.815	0.834	0.731
	LightGBM	0.814	0.817	0.708
	Random Forest	0.835	0.834	0.735
	CapsNet	0.816	0.810	0.717
	Transformer	0.829	0.825	0.702
	Transformer-CapsNet	0.828	0.819	0.723
	TabFM	0.849	0.847	0.748
Clinical-only	Gradient Boosting	0.799	0.785	0.628
	LightGBM	0.774	0.740	0.639
	Random Forest	0.790	0.771	0.634
	CapsNet	0.794	0.776	0.638
	Transformer	0.809	0.770	0.615
	Transformer-CapsNet	0.800	0.786	0.641
	TabFM	0.803	0.797	0.635
Radiomics+clinical (concat)	Gradient Boosting	0.843	0.861	0.717
	LightGBM	0.833	0.858	0.729
	Random Forest	0.850	0.863	0.725
	CapsNet	0.843	0.829	0.742
	Transformer	0.859	0.853	0.704
	Transformer-CapsNet	0.841	0.841	0.708
	TabFM	0.872	0.874	0.734
Radiomics+clinical (dual)	Gradient Boosting	0.843	0.861	0.717
	LightGBM	0.833	0.858	0.729
	Random Forest	0.850	0.863	0.725
	Dual CapsNet	0.857	0.857	0.697
	Dual Transformer	0.860	0.855	0.706
	Dual Transformer-CapsNet	0.842	0.842	0.699
	TabFM (late fusion)	0.874	0.870	0.743